

Schmitt Trigger Oscillator

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Design:

For this Design Problem, I decided to go with an alternative option to design a circuit of my choosing. I went with the Schmitt Trigger Oscillator, a square wave generator that consists of a schmitt trigger and an RC feedback network as shown in Figure 1. Essentially, the schmitt trigger takes advantage of hysteresis and feeds an input signal back to itself. Based on the trigger threshold levels, the duty cycle of the output square wave can be modulated. Additionally, the frequency of the output waveform can be modulated relative to the RC time constant of the feedback network.

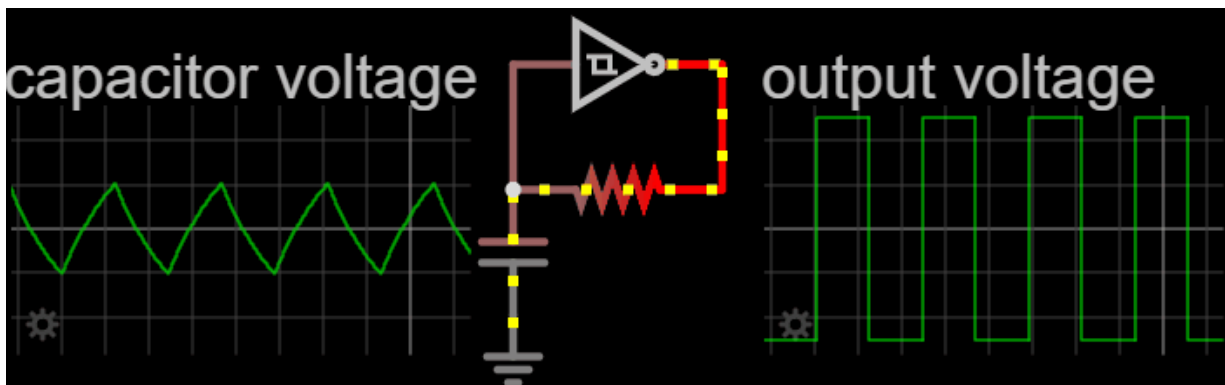


Figure 1: Schmitt Trigger Oscillator Topology using a feedback resistor and capacitor across an inverting schmitt trigger

The schmitt trigger I chose followed the topology described by Filanovsky et al. in which an inverter supported by transistors between the rails contains an additional set of feedback transistors. Based on the ratio of the widths of the feedback transistor to its respective nmos or pmos, the threshold level of the trigger can be modulated. According to Filanovsky et al., the max bandwidth between the thresholds is about 0.2V. I found this out to be true through experimentation as setting the width ratio to the max that Cadence could handle maintained the 0.2V bandwidth. I landed on a ratio between widths of 1 for the pmos' and 0.6 for the nmos' to produce thresholds of 660 mV and 7.9 mV. The resulting schmitt trigger is shown in Figure 2.

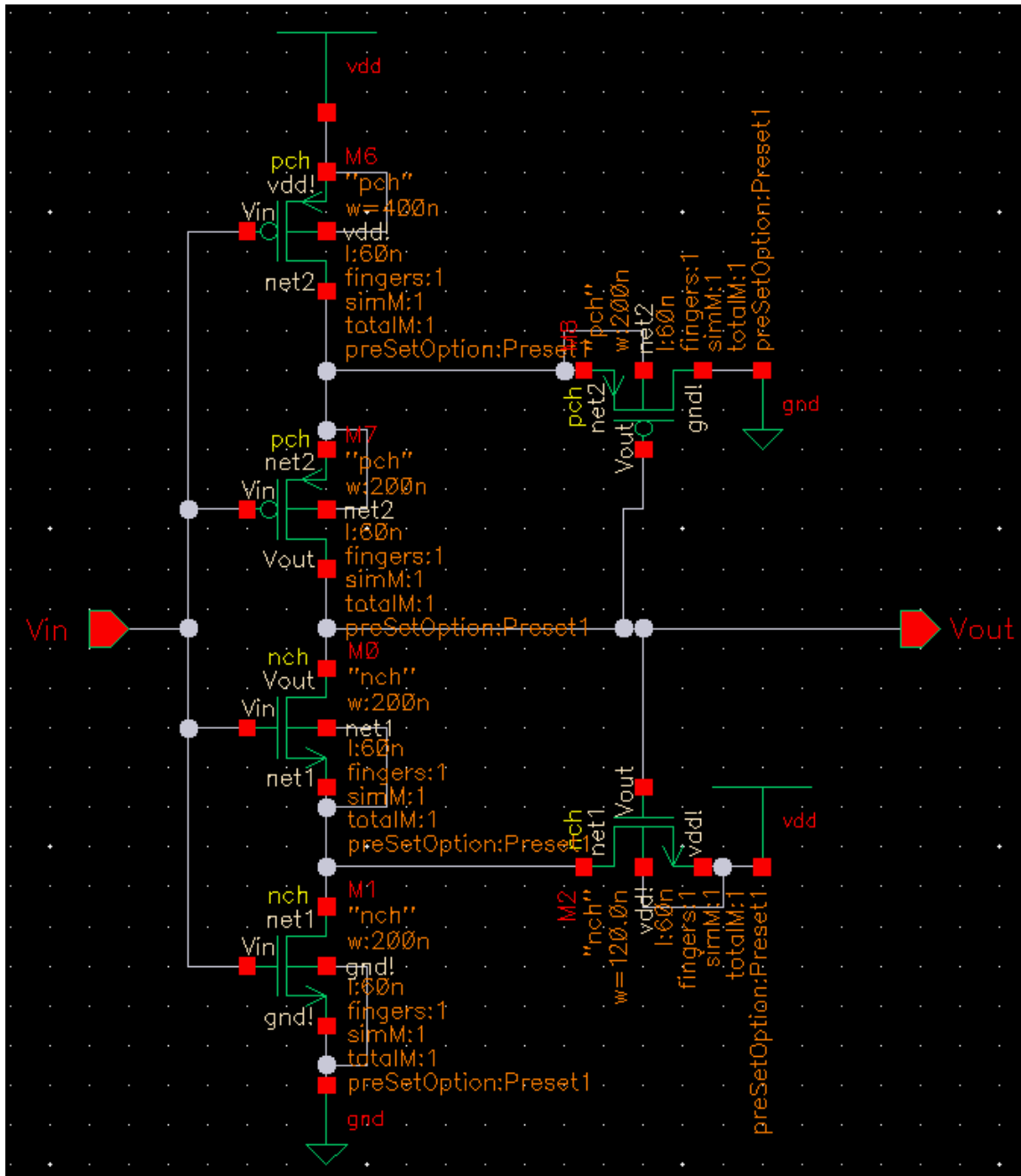


Figure 2: Schmitt Trigger circuit in Cadence. The widths of the feedback transistors M2 and M8 relative to M0 and M7 respectively control the high and low threshold voltages

To produce an output wave of sufficiently low enough frequency, a relatively high value resistor and capacitor pair are required. For the 65nm process, it would be inefficient to create a resistor of excessive magnitude such as 100 kOhms. Thus, I decided to include a Switched Capacitor (SC) circuit to take the place of the resistor. My initial design consisted of two pass n-channel mosfets with a capacitor in between. These transistors were fed opposite high frequency square

waves to output a resistance described by the equation: $R = \frac{1}{C_s * F_s}$. Upon simulation of the feedback network using the SC and a feedback capacitor, I found the resulting wave to have a much higher rise time than fall time. Thus, I added a parallel set of p-channel transistors to the existing n-channel transistors to offset their degradation properties. The revised SC, shown in Figure 3 made a balanced input waveform for the schmitt trigger.

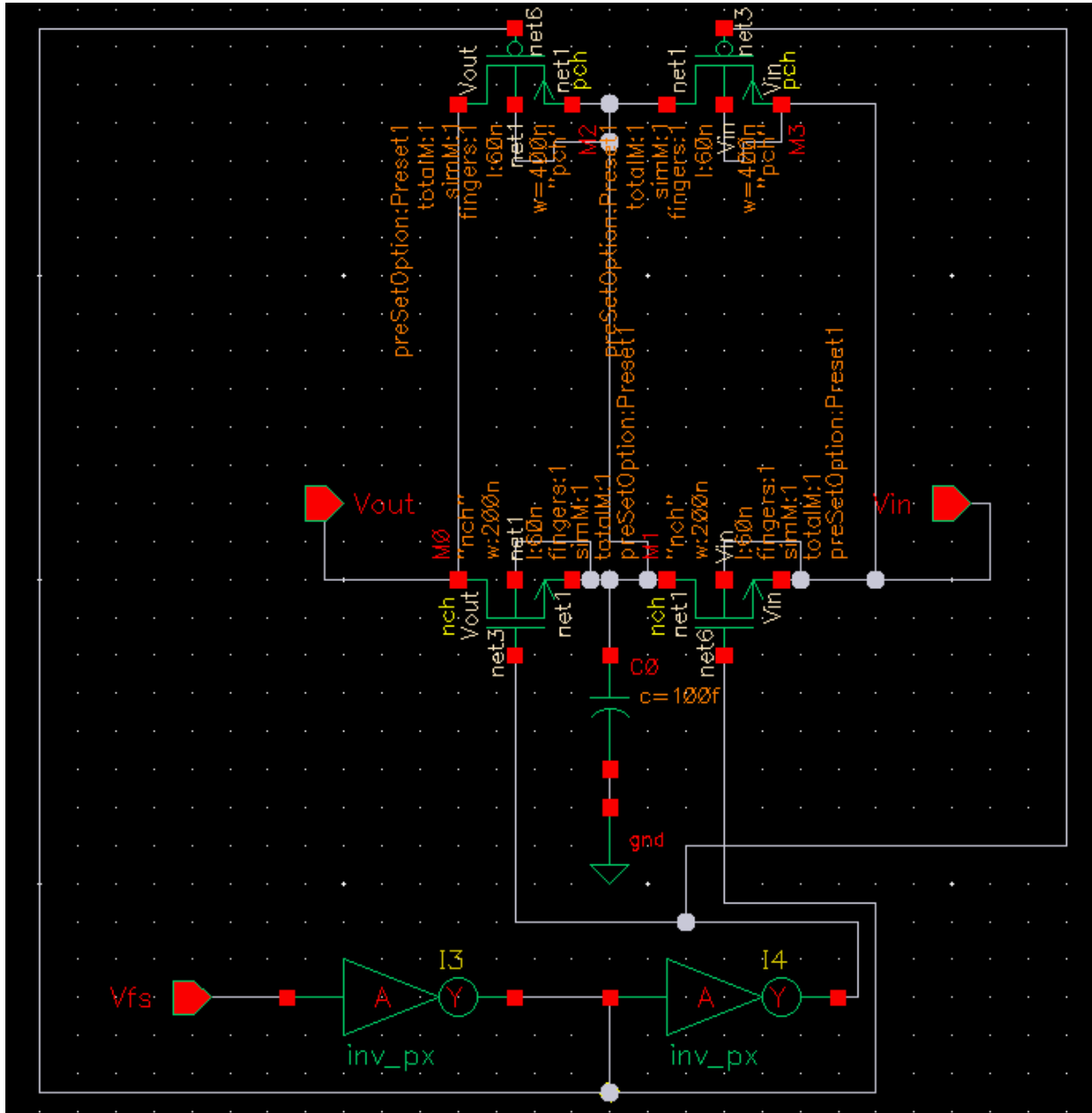


Figure 3: Switched Capacitor circuit with parallel pass transistor branches and a switch capacitor of 100f.

Lastly, to handle the requirements of the 10f-10p load, an inverter chain was also designed. Assuming an input capacitance of 12 fF and a load capacitance of 10 pF, it was found that a chain of 10 transistors with a multiplier of 4 would be needed to support the high load. The resulting chain is shown in Figure 4.

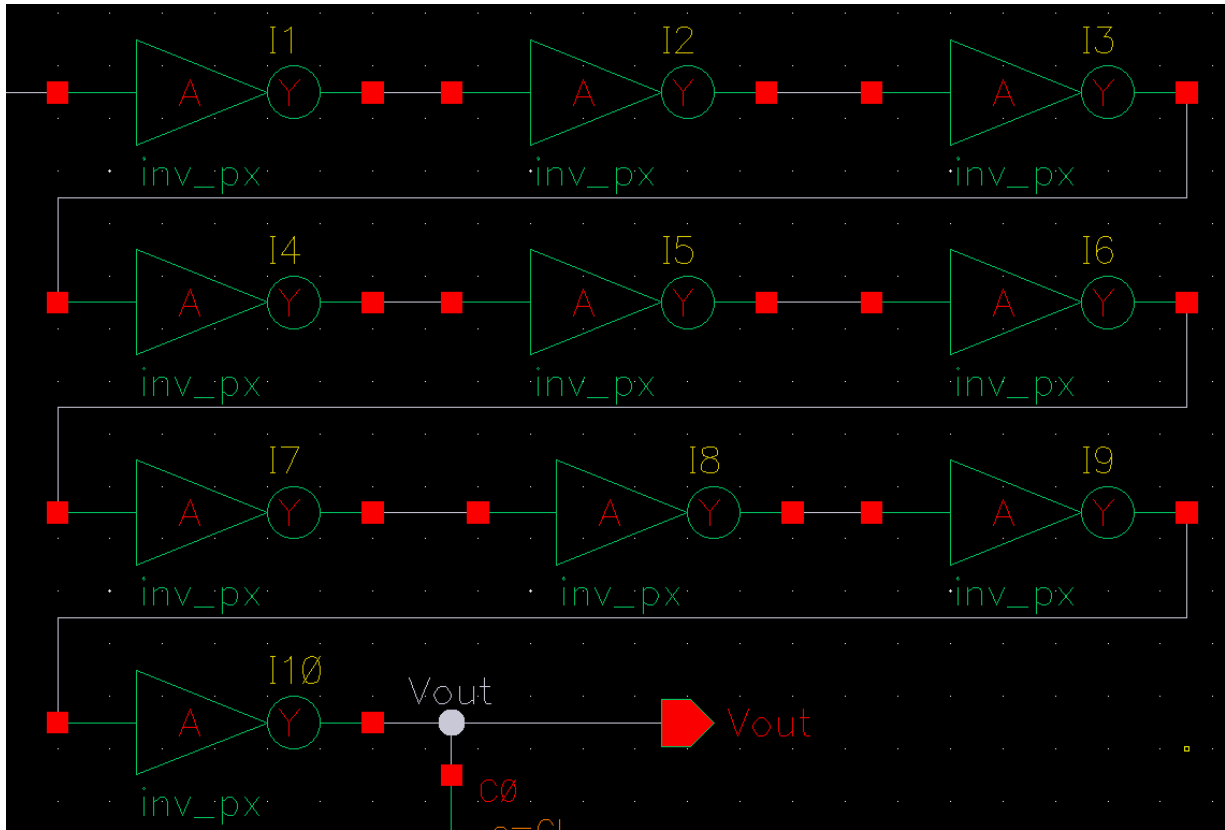


Figure 4: Inverter Chain connecting to the Vout pin and load capacitor.

Simulation:

For simulation, I took the final circuit shown in Figure 5 and applied a load capacitor on the output of the inverter chain. Additionally, for dc analysis, the Vdd value was swept from 0-1.2 using the variable “vpwr”; this was especially useful when testing the schmitt trigger individually as the hysteresis sweep helps visualize the component’s thresholds seen in Figure 6. For the transient response, the stop time was set to 2 ms as the expected waveform frequency was set for 20 kHz. The results of the dc analysis are shown in Figure 7 while the resulting transient analysis plot is shown in Figure 8. The dc analysis shows the current draw as Vin increases. The current draw reaches a maximum of 7.48 fA at 370 mV. The transient analysis shows that the output waveform is successfully a square wave with a frequency of 19.25 kHz. A plot of the calculated duty cycle was overlaid showing a consistent duty cycle of 49.8%.

Lastly, the impact of the load sizes can be seen in Figure 9. Even with the high load, the wave form stayed around 14 ps and 12 ps in rise and fall times respectively.

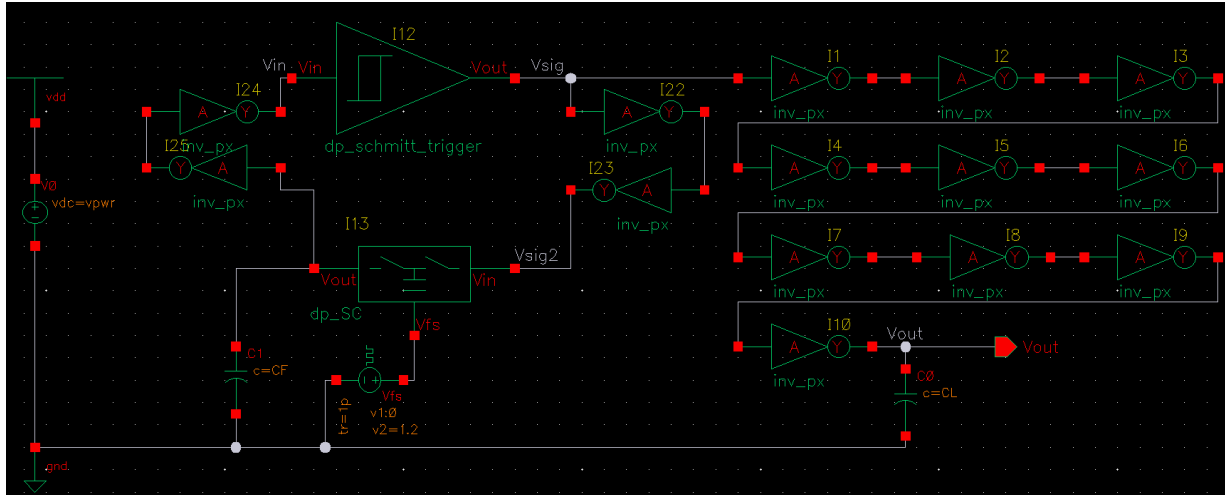


Figure 5: Full circuit schematic. The Schmitt trigger component has buffers on both sides with the SC circuit and capacitor CF acting as the feedback components. V0 sets the Vdd of the circuit while Vfs sets the switching frequency of the SC component.

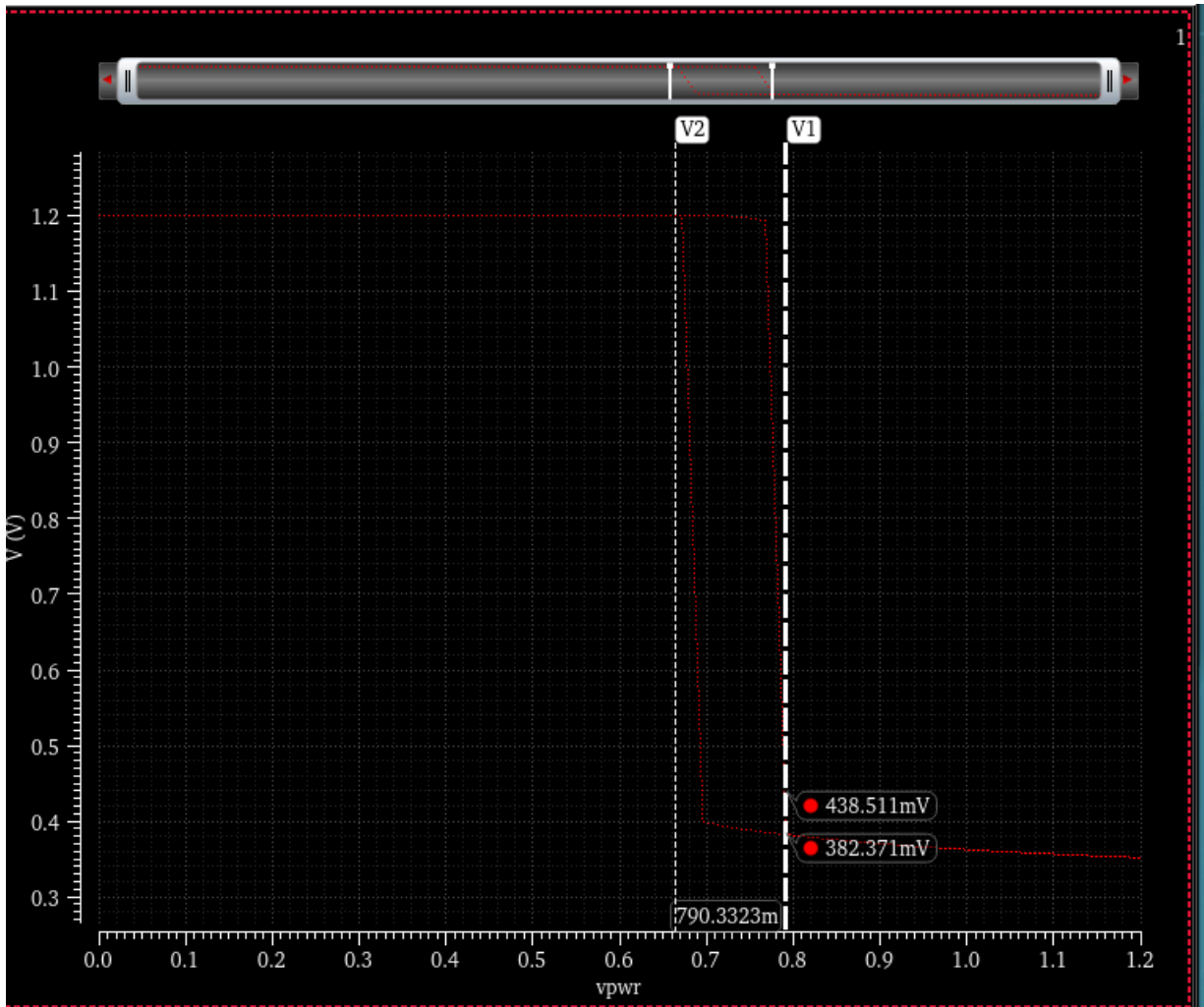


Figure 6: Hysteresis curve for Schmitt trigger. Low threshold of 660 mV and High threshold of 790 mV.

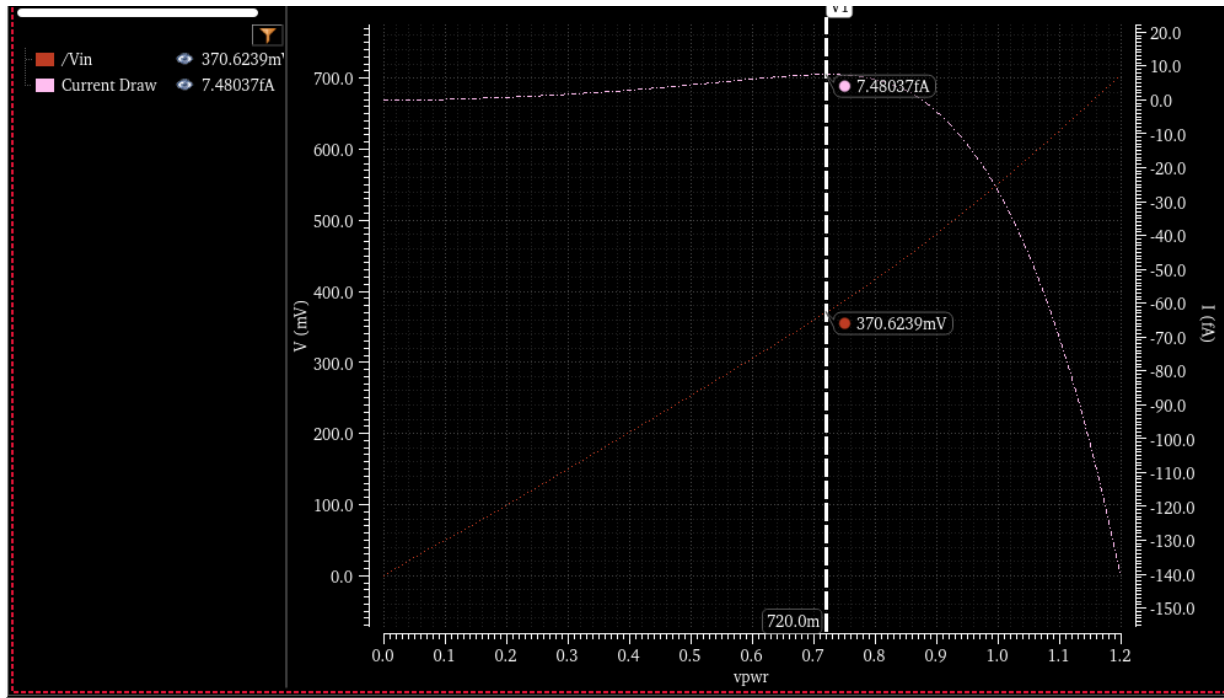


Figure 7: DC Response Plot. Peak current draw seen at a V_{in} of 370 mV was 7.48 fA .

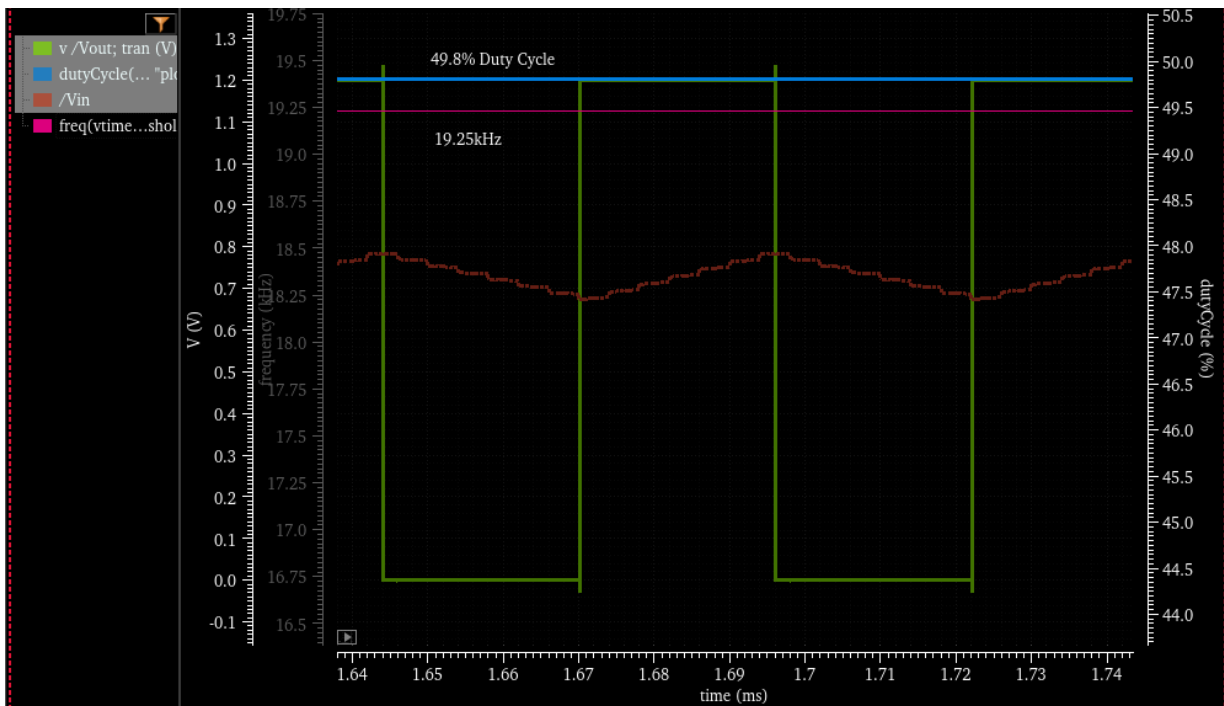


Figure 8: Transient Response Plot at a load of 10 pF . A duty cycle of 49.8% and a frequency of 19.25% were measured by the Cadence calculator

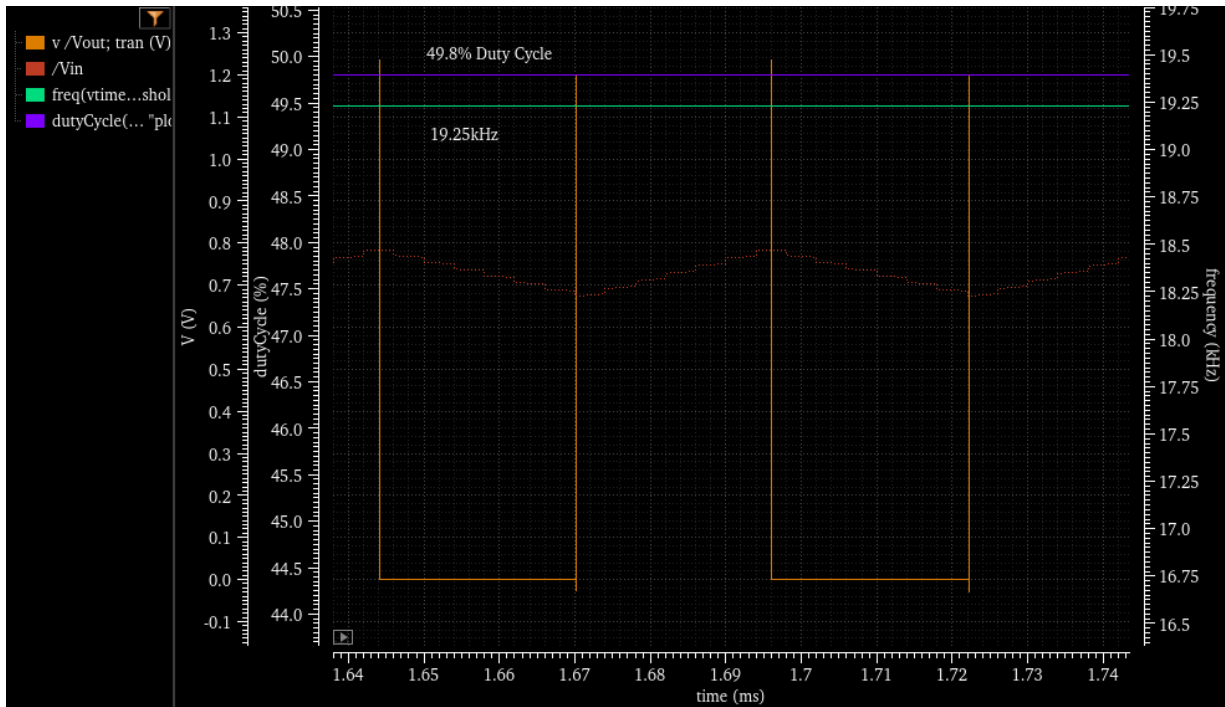


Figure 9: Transient Response plot at a load of 10 fF comparing the effects of the load capacitor. Minimal differences seen in duty cycle or frequency

Specifications:

Performance Objective	Simulated Values
INPUT Signal: 250 kHz Square Wave	250 kHz Square Wave
OUTPUT Signal: 20 kHz Square Wave	19.25 kHz Square Wave
OUTPUT Duty Cycle: 50% +/- 1%	49.8%
OUTPUT Start Up Time: <10us	< 0.2 μ s
OUTPUT Rise/Fall Time: <10ns	14 ps & 12 ps
Schmitt Trigger Low Threshold: 0.6 V +/- 0.1 V	0.67 V
Schmitt Trigger High Threshold: 0.8 V +/- 0.1 V	0.78 V
OUTPUT Low and High Values: < 100mV & > 1.1V	130 nV & 1.2 V
DC Current Draw with INPUT at VOL & VOH: < 5uA	< 732 nA

Layout

For the layout seen in Figure 10, the original circuit was adjusted to facilitate the amount of effort and placement needed. The inverter chain was reduced from 10 to 2 as the multiplier would lead to an excessive amount of transistors. All capacitors were replaced with CRTMOM equivalents close to the original values listed in the schematic. More detailed images of the layout can be found in Appendix B.

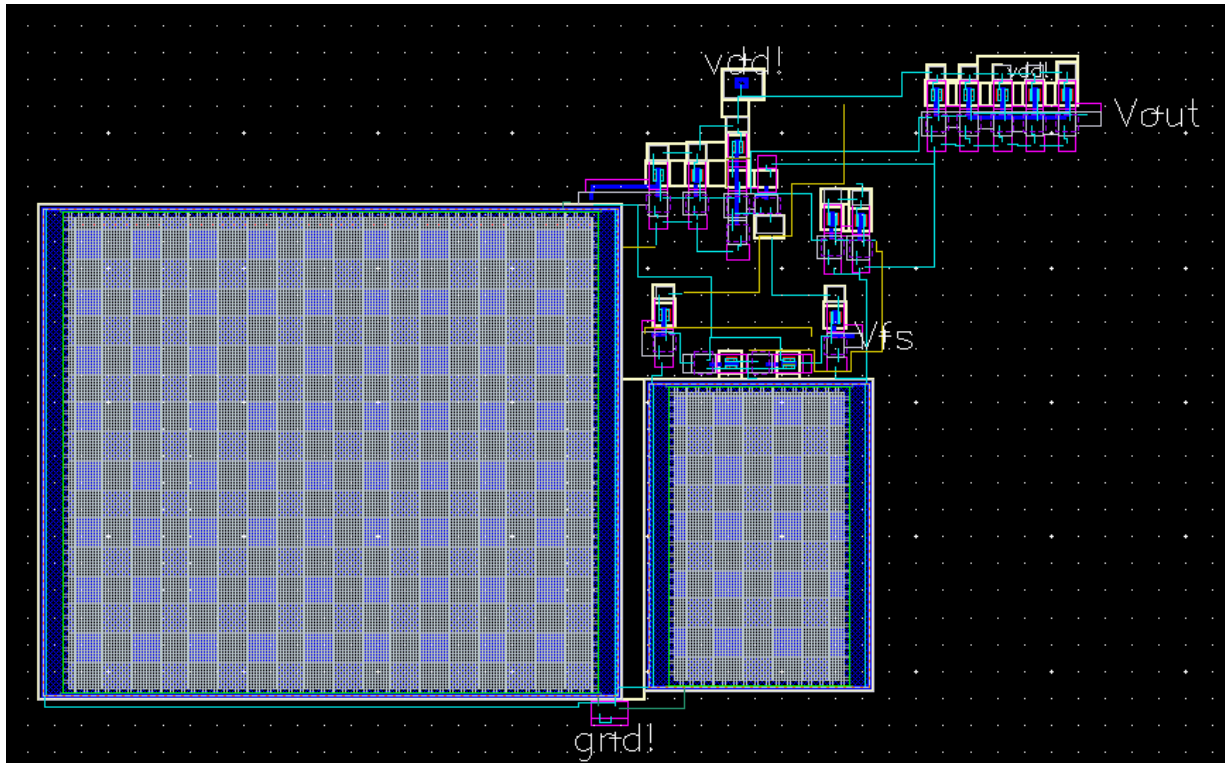


Figure 10: Overall Circuit Layout

Running the DRC resulted in the report shown in Figure 11. The only issues that remained were Latch Up errors which are negligible for this process.

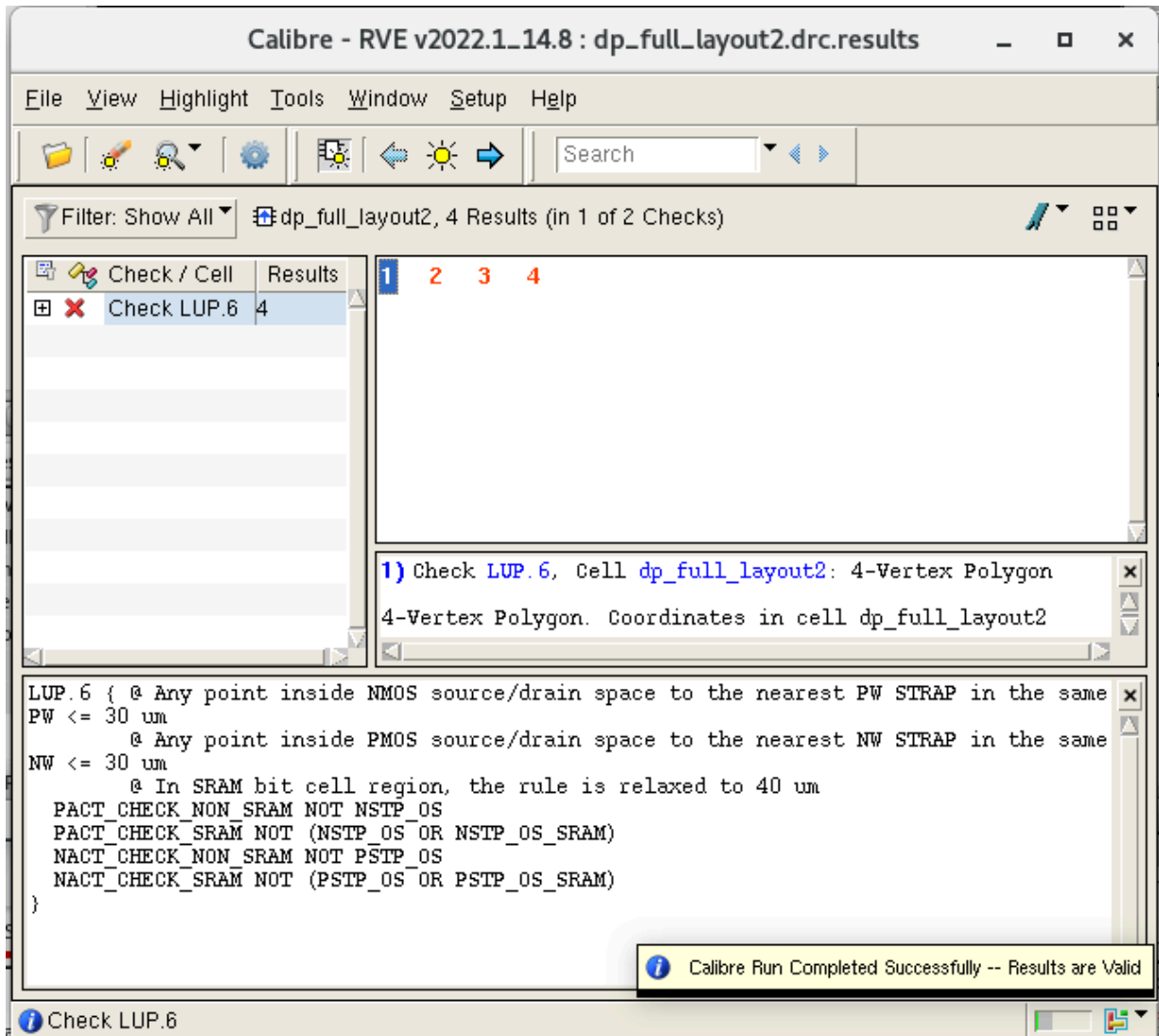
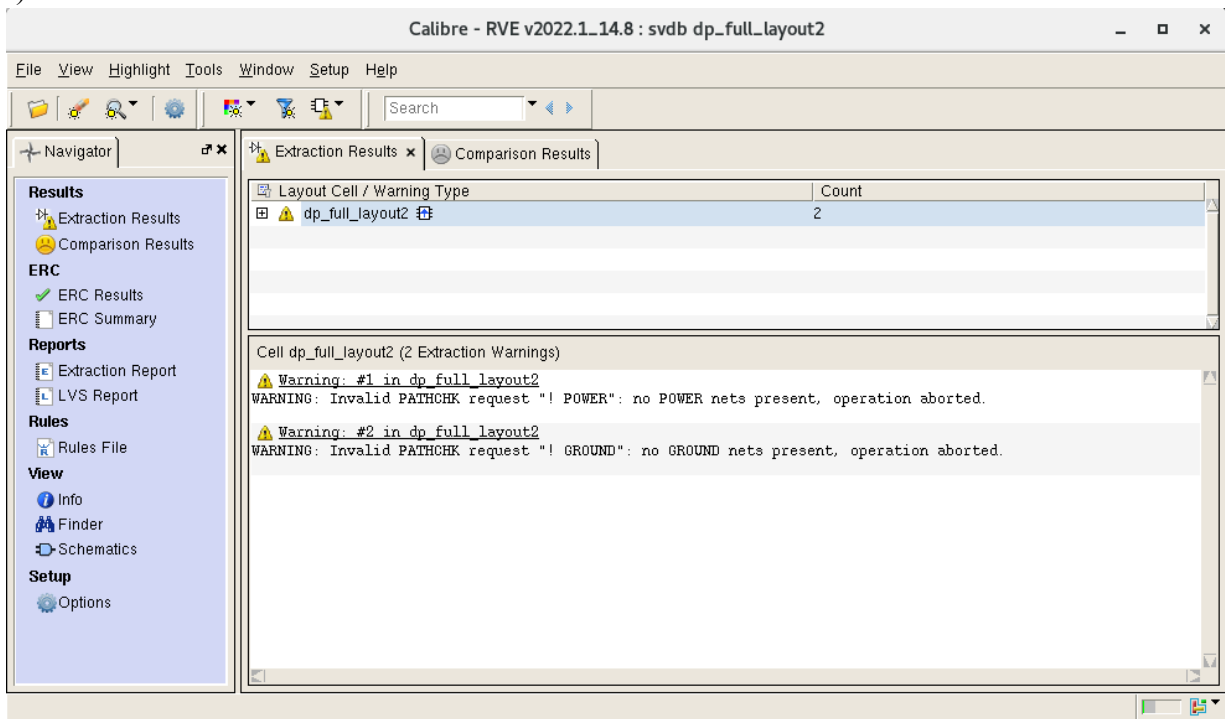


Figure 11: DRC Report with only 4 LUP errors shown

Lastly, the LVS report in Figure 12 shows that the layout circuit output did not match the schematic. Due to an untraced connection, port, or net error, the LVS was prevented from running to a successful completion. Despite thorough examination of the layout and fixing a couple of unconnected traces, the LVS error count failed to decrease. With more time and effort, the circuit may be made ready for fabrication, but in its current state it is not.

a)



b)

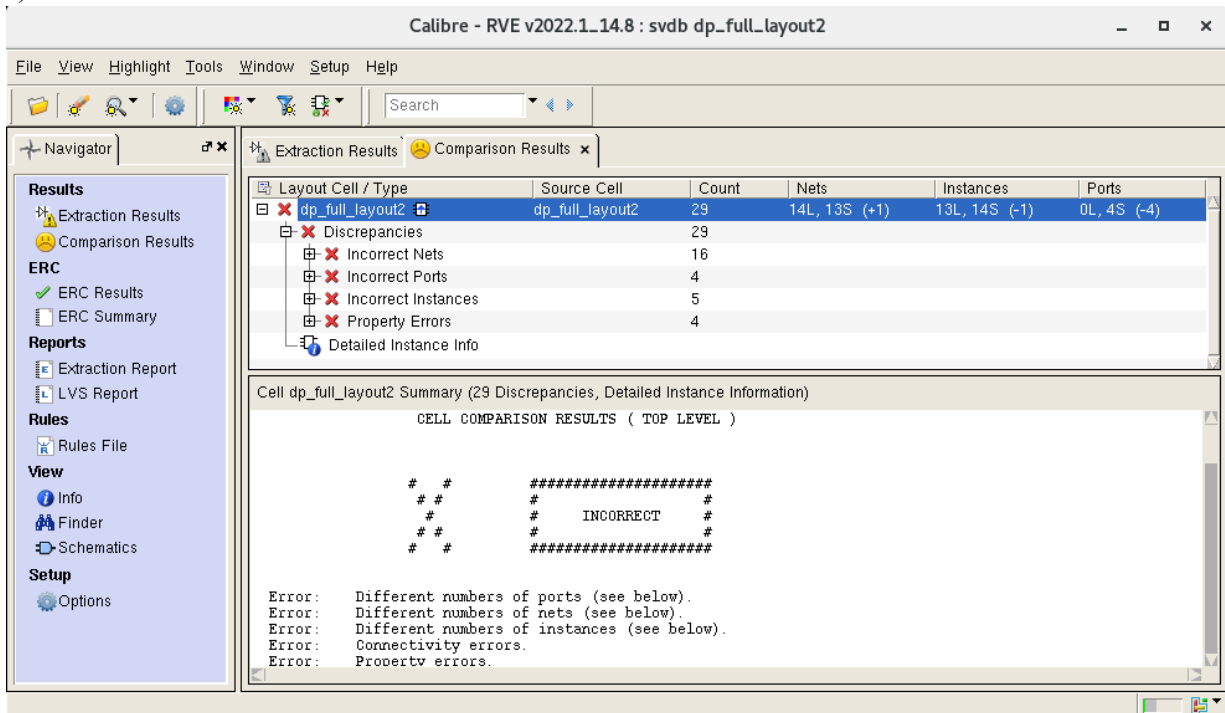


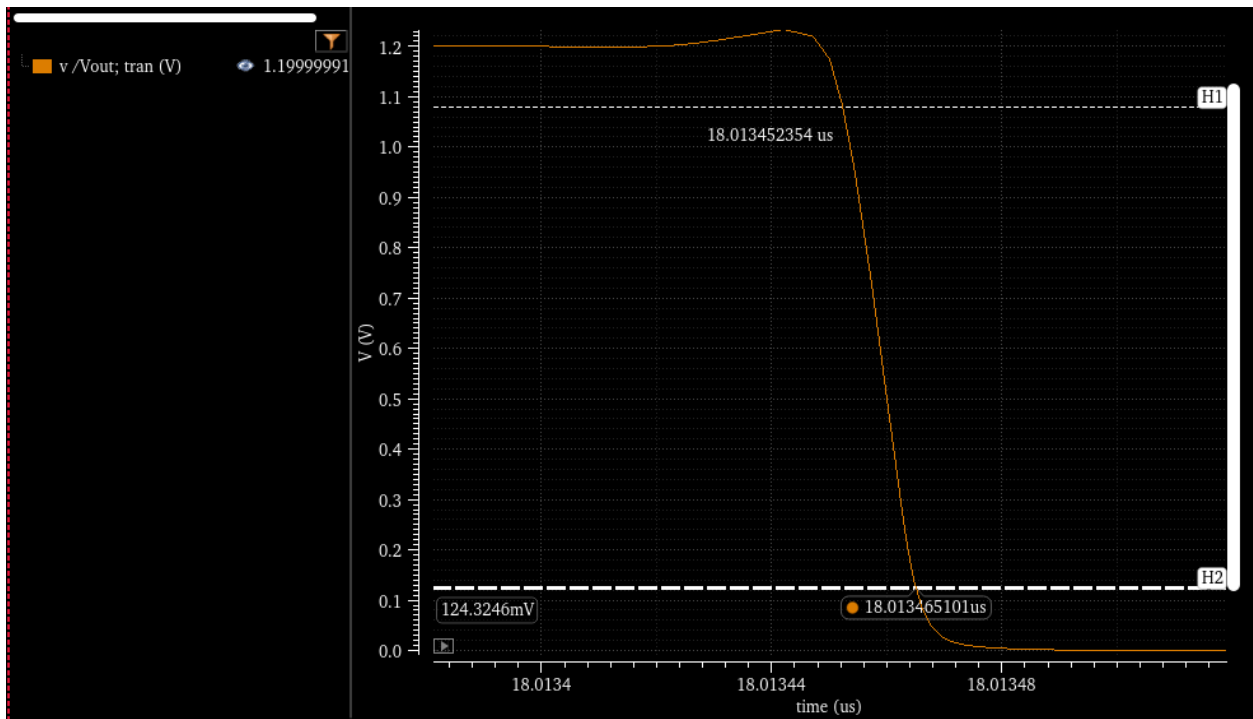
Figure 12: LVS Report a) Extration Results and b) Comparison Results

Works Cited:

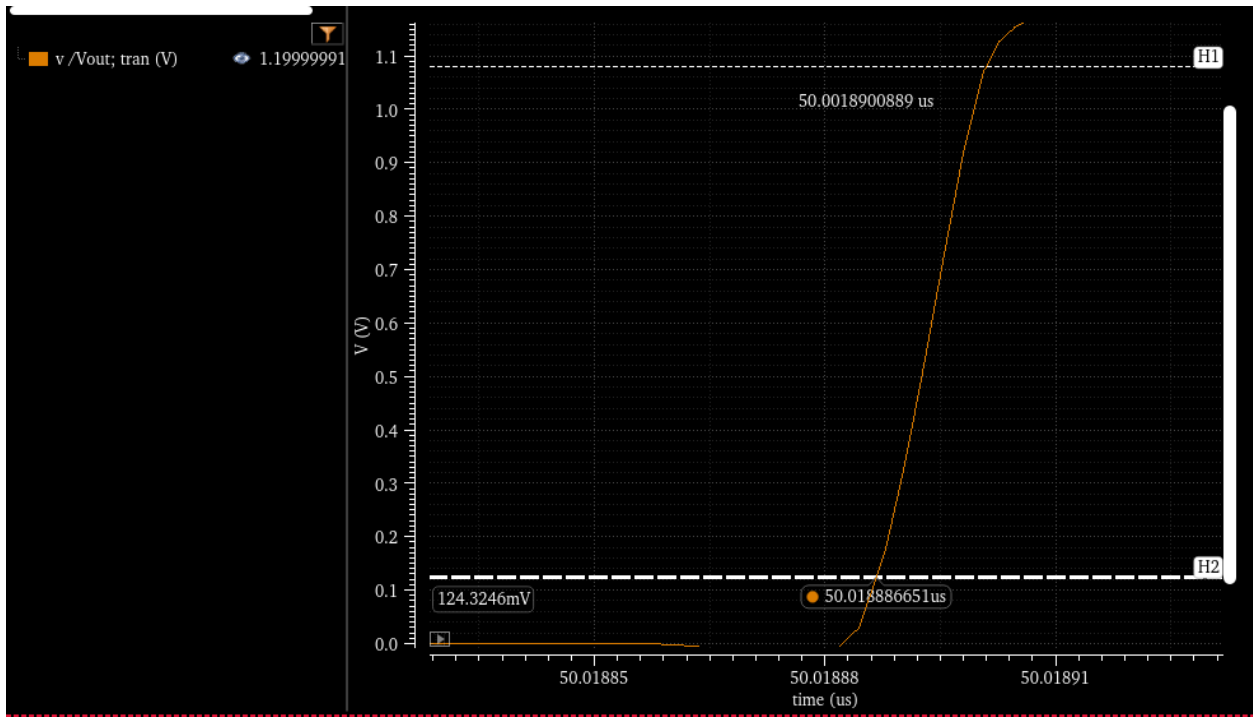
Filanovsky, I. M., & Baltes, H. (1994). CMOS Schmitt trigger design. IEEE Transactions on Circuits and Systems-I: Fundamental Theory and Applications, 41(1), 46-49.

Appendix A: Additional Plots

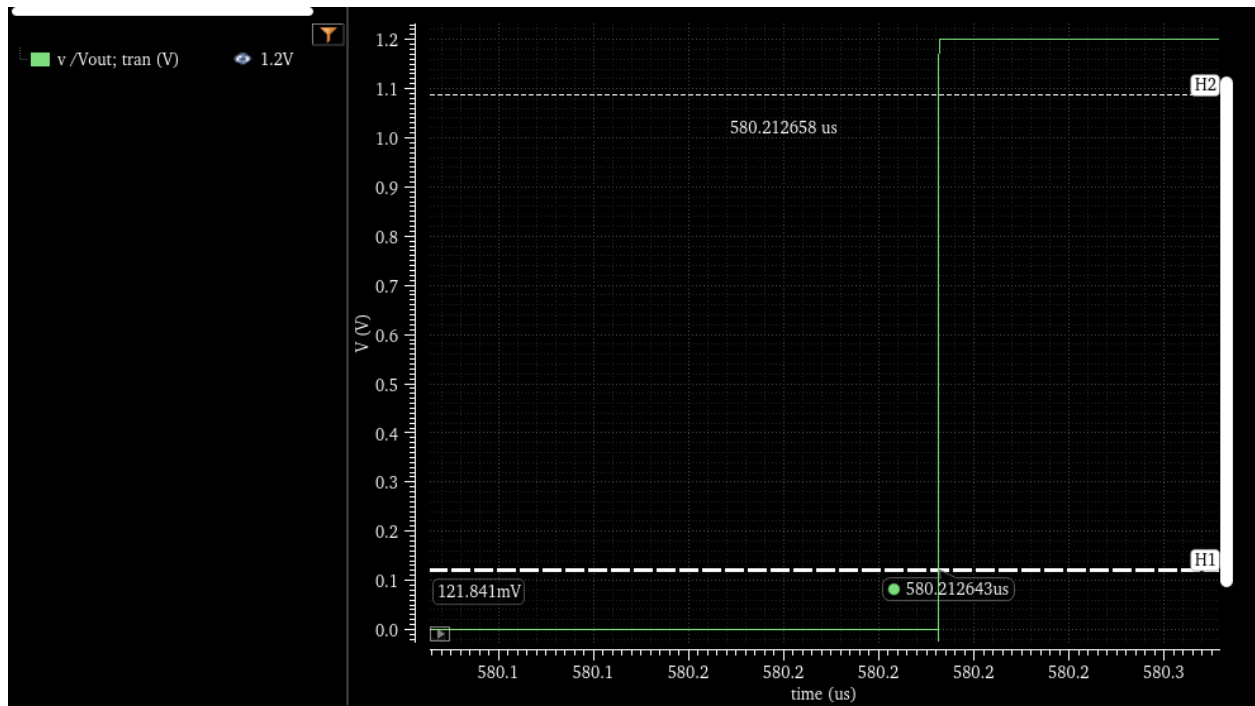
Fall Time Plot 10 fF Load



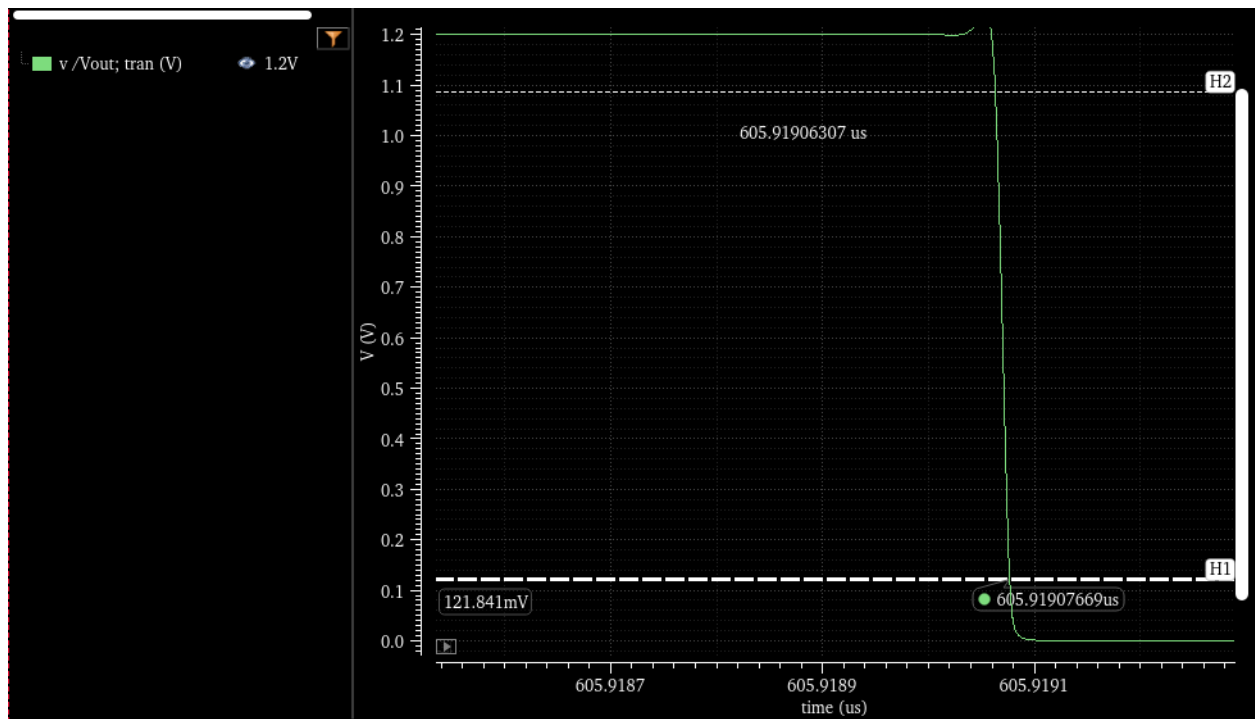
Rise Time Plot 10 fF Load



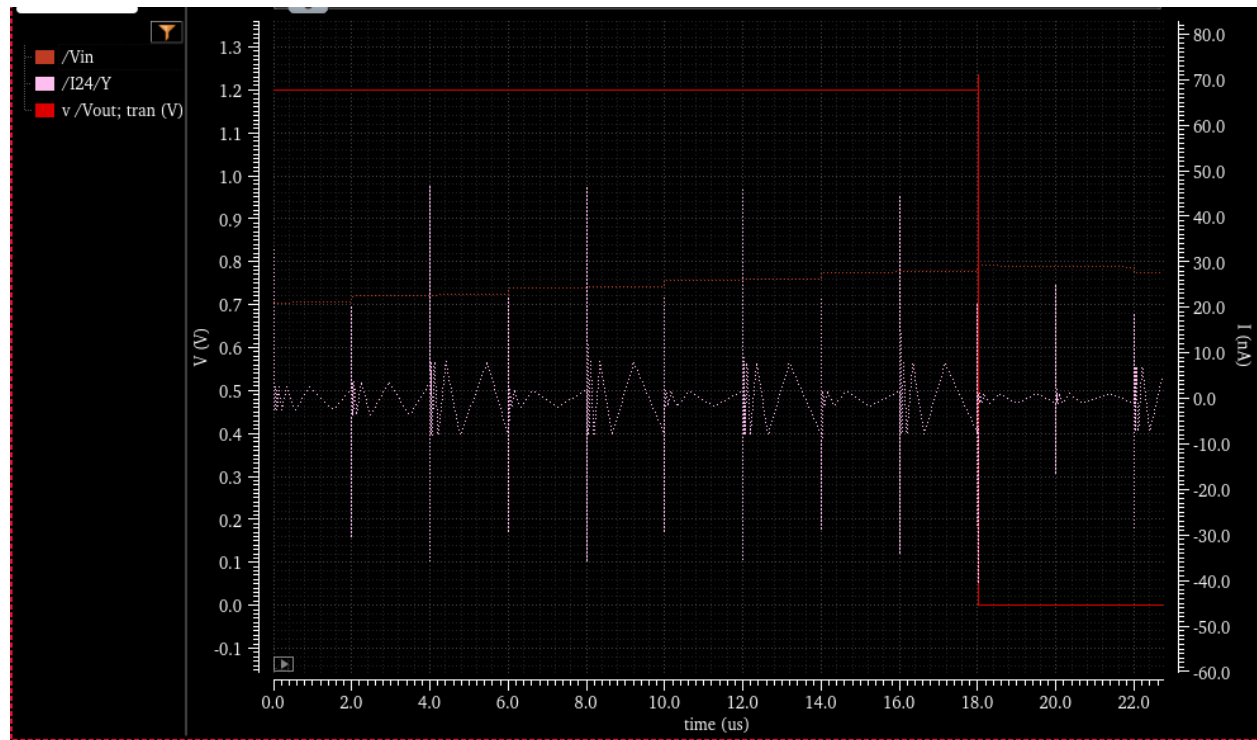
Rise Time Plot 10 pF Load



Fall Time Plot 10 pF Load

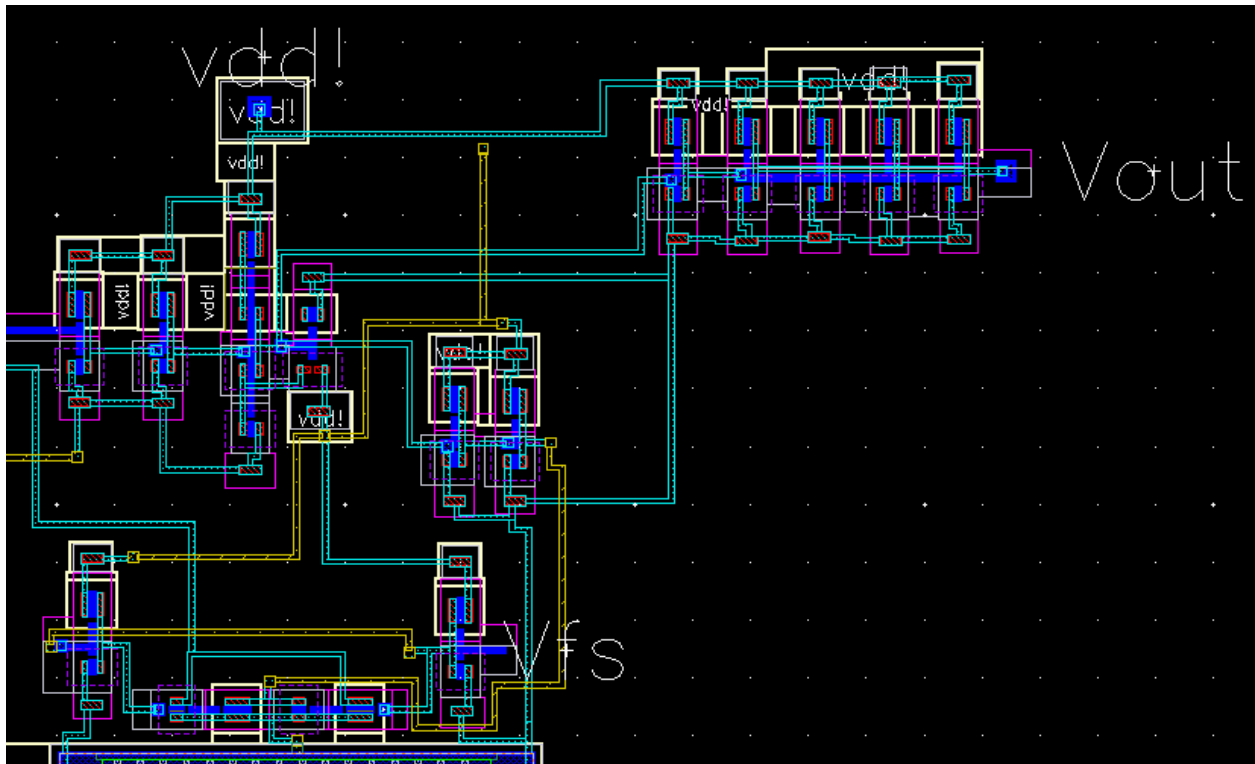


Plot showing Start Up Time:

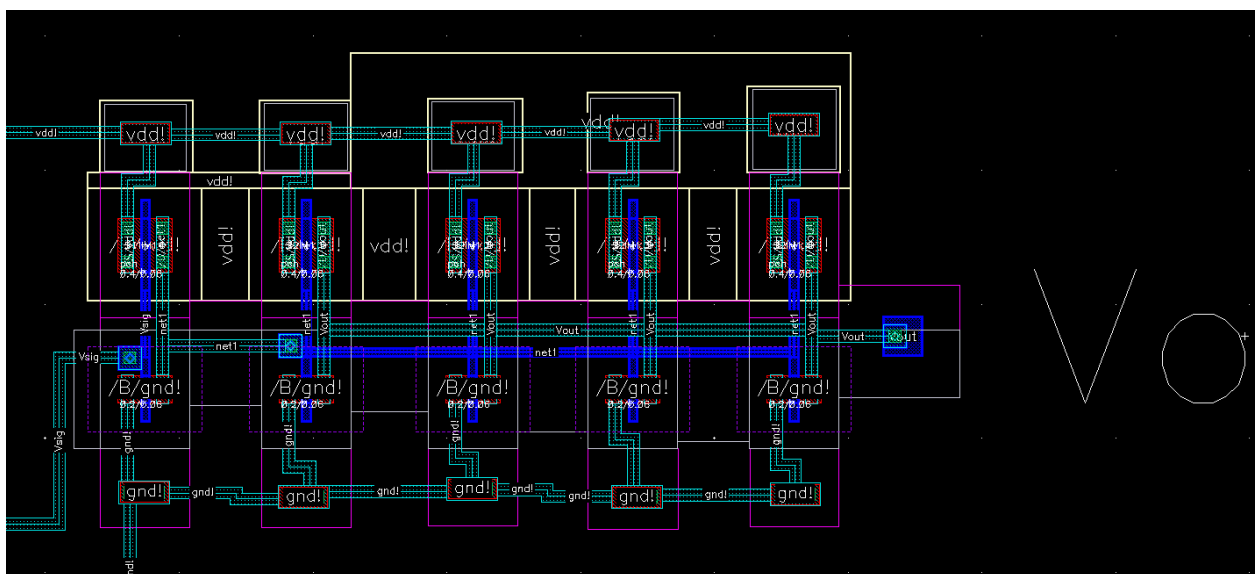


Appendix B: Additional Layout Images

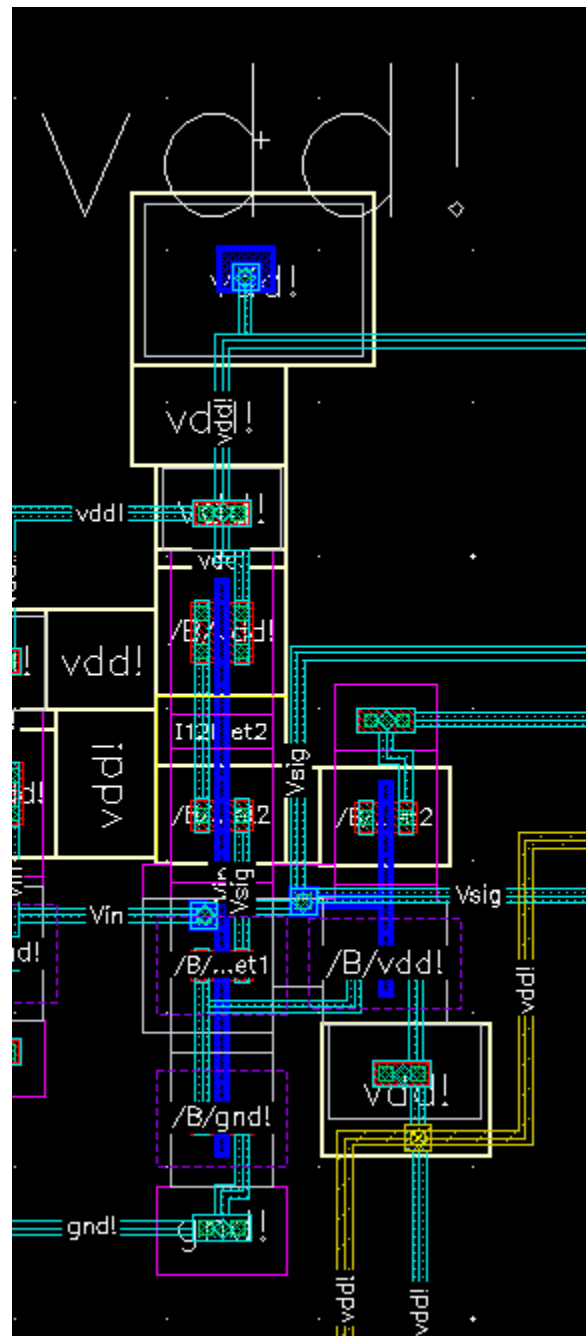
Layout Exlcuding the Capacitors:



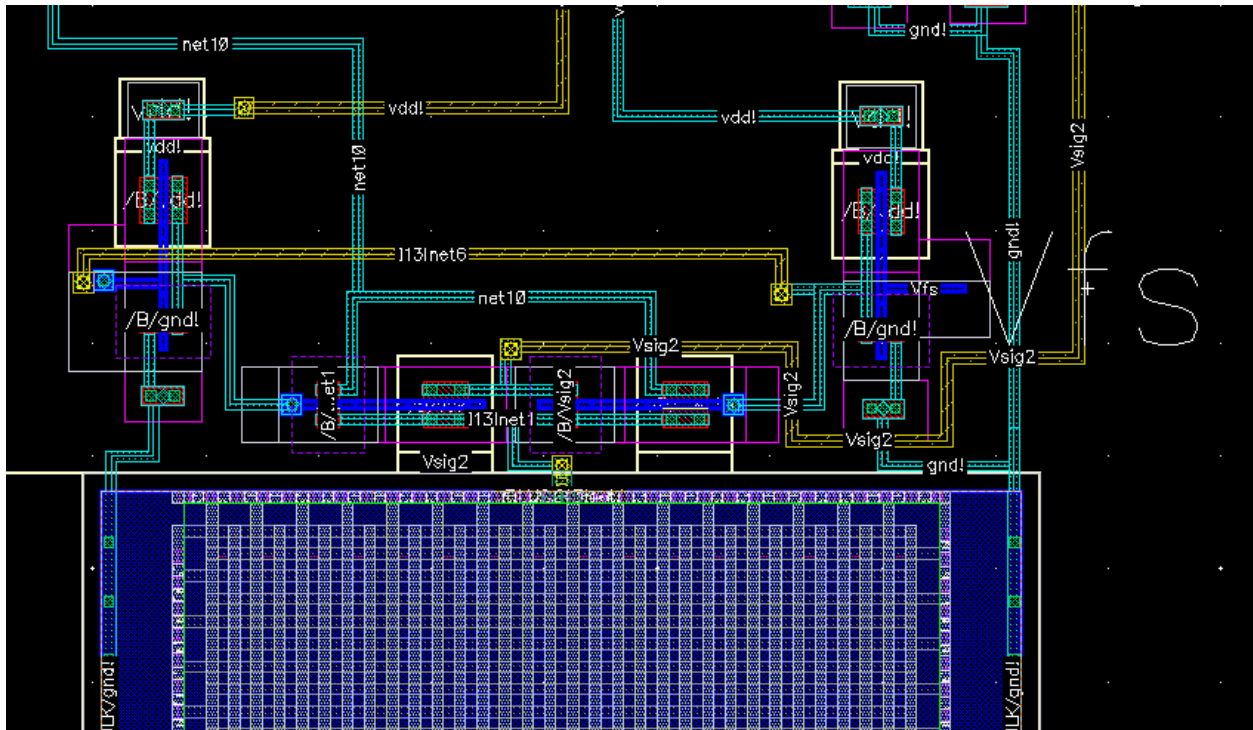
Layout of Inverter Chain:



Layout of Schmitt Trigger:



Layout of Switched Capacitor:



Layout of Buffer:

